denote the electric field, the magnetic field, the electric charge density, the electric current density, the permittivity of free space, and the permeability of free space, respectively. As you can see, Maxwell's equations relate the rate of change of the electromagnetic fields to the presence of electric charges and currents. It is not hard to show that these equations imply a speed for electromagnetic waves given by $1/\sqrt{\varepsilon_0 \mu}$, which when evaluated is in fact the speed of light.

2. There is some controversy as to the role such experiments played in Einstein's development of special relativity. In his biography of Einstein, *Subtle Is the Lord: The Science and the Life of Albert Einstein* (Oxford: Oxford University Press, 1982), pp. 115–19, Abraham Pais has argued, using Einstein's own statements from his later years, that Einstein was aware of the Michelson-Morley results. Albrecht Fölsing in *Albert Einstein: A Biography* (New York: Viking, 1997), pp. 217–20, also argues that Einstein was aware of the Michelson-Morley result, as well as earlier experimental null results in searching for evidence of the aether, such as the work of Armand Fizeau. But Fölsing and many other historians of science have also argued that such experiments played, at best, a secondary role in Einstein's thinking. Einstein was primarily guided by considerations of mathematical symmetry, simplicity, and an uncanny physical intuition.

3. For us to see anything, light has to travel to our eyes; similarly, for us to see light, the light itself would have to make the same journey. So, when I speak of Bart's seeing light that is speeding away, it is shorthand. I am imagining that Bart has a small army of helpers, all moving at Bart's speed, but situated at various distances along the path that he and the light beam follow. These helpers give Bart updates on how far ahead the light has sped and the time at which the light reached such distant locations. Then, on the basis of this information, Bart can calculate how fast the light is speeding away from him.

4. There are many elementary mathematical derivations of Einstein's insights on space and time arising from special relativity. If you are interested, you can, for example, take a look at Chapter 2 of *The Elegant Universe* (together with mathematical details given in the endnotes to that chapter). A more technical but extremely lucid account is Edwin Taylor and John Archibald Wheeler, *Spacetime Physics: Introduction to Special Relativity* (New York, W. H. Freeman & Co., 1992).